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A NOVEL INTERVAL-HALVING ALGORITHM FOR PROCESS TREND IDENTIFICATION

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Abstract: Qualitative process trend representation is an useful approach to model the temporal evolution of sensor data and has been applied in areas such as process monitoring, data compression and fault diagnosis. However, the sheer volume of real-time sensor data that needs to be processed necessitates an *automated* approach. The first step of recovering the important temporal features is made difficult because of the absence of *a priori* knowledge about trend characteristics such as noise and varying scales of evolution. In this paper, we propose a novel approach to automatically identify the qualitative shapes of trends using an interval-halving based polynomial-fit technique. To estimate the significance of fit-error, wavelet-based denoising is used. The procedure identifies piecewise unimodals represented as quadratic segments. Finally, a unique assignment of qualitative shape is made to each of the identified segments. The application of the technique is illustrated on both simulated and industrial data. *Copyright© 2001 IFAC*

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1. INTRODUCTION

Process data contain valuable information about the state, operation and behavior of the concerned systems, more so in cases with limited available process knowledge i.e., 'data rich and information poor' systems. Increased automation and faster sampling rates have made large volumes of precise data collection and rapid access from electronic devices possible (Kennedy, 1993). The problem of interpretation i.e., extraction of meaningful information, however, has largely been left to the operator, who usually suffers from information overload. The potential uses (Mah *et al.*, 1992; Rengaswamy *et al.*, 2001) of data are numerous: detection and diagnosis of faults, adaptive control, product quality control and operator training to name a few. This motivates development of methods which extract useful and relevant information from sensor data.

Process trend analysis is an useful approach to exploit the temporal information and reason about process state. The main activities involved are (1) A technique to *identify* trends and (2) A *mapping* from trends to operational conditions. While formal trend representation schemes have been proposed, the question of *automatically* and *correctly* identifying such features is still a challenging task. There is a need for a robust, accurate and a simple yet effective scheme for real-time interpretation of trends to allow for adequate temporal reasoning mechanisms. We will use the primitive-

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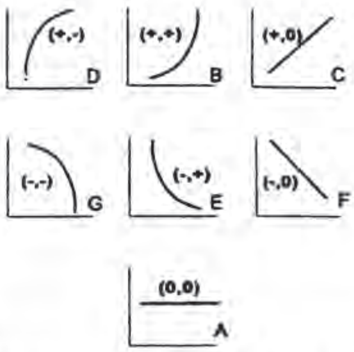

Fig. 1. Fundamental Language: Primitives

based language (Rengaswamy and Venkatasubramanian, 1995) in this paper. The trend profiles are composed of fundamental elements called primitives (Figure 1).

The organization of the paper is as follows: the main issues in trend identification are outlined in the next section. Section 3 briefly reviews the current techniques and presents the new idea. Wavelet-based-denoising is also discussed there. In Section 4 we propose the novel approach to automate trend identification based on an *interval-halving* procedure. We illustrate its application in Section 5 to elucidate the effectiveness of the technique to simulated and real-data.

2. ISSUES IN PROCESS TREND IDENTIFICATION

Some of the main issues involved in the trend identification process are elaborated upon below:

(1) **Time-Scale of Identification:** The temporal behavior of measured variables in a chemical process is the superposition of many underlying driving processes occurring at different time-scales (Bakshi and Stephanopoulos, 1994). Hence the *scales* at which real trends evolve may vary considerably from slow (equipment degradation) to extremely fast (stuck valves). The problem of deciding the width of the time window (window size) for trend-identification is a very crucial problem. Choosing either a too narrow time window or a large one can lead to erroneous identification.

(2) **Sensor Noise:** Sensor data are invariably corrupted with measurement noise. The signal to noise (S/N) ratio is an important criteria in deciding the trend-analysis amenability of a sensor. If this ratio is ≤ 1 (more noise than signal), the data may not carry much trend information. To be able to evaluate this, one needs to estimate the 'true' i.e., *denoised* data. Wavelets are a powerful tool for denoising with minimum *a priori* assumptions. This is discussed in detail in Section 3.1.

(3) **Simplicity and Computational complexity:** Trends have a 'visual' feel and are simple to relate to. As such, it is desirable to have simplicity in extraction and reasoning while not compromising on accuracy and applicability. Also, since most of the intended applications are real-time, the computational complexity of the algorithm should not be prohibitive enough to restrict its usefulness.

3. BRIEF REVIEW AND BASIC IDEA

Brief Review: Recognizing the usefulness and potential of temporal behavior represented by process trends has led to the development of formal representations and mechanisms for identification of the same. A formal qualitative representation scheme consisting of 'triangular' episodes (Cheung and Stephanolpoulos, 1990a) was used to extract trends by Bakshi and Stephanopoulos (1994) using multiscale wavelet analysis. Konstantinov and Yoshida (1992) approximated the noisy process signal by a polynomial but the time-scale of analysis was fixed *a priori*. Janusz and Venkatasubramanian (1991) developed a 'minimal' trend language whose fundamental elements called primitives (Figure 1) were generated using a finite difference method. Later, this work was extended using a syntactic pattern recognition (Rengaswamy and Venkatasubramanian, 1995) approach where the primitives were identified using a fixed-size neural network. Vedam and Venkatasubramanian (1997) proposed an adaptive trend analysis system based on wavelet theory where data was projected onto the scaling function at different levels and the coefficients were fed to a neural network to identify the primitives. The problems here were the size of the input layer in the network, neural network training and noise. Also the whole process being NN-based is not transparent.

Basic Idea: In order to facilitate conversion of numerical data into a symbolic form we associate a functional form (polynomial) to the data. Note that the shapes of the primitives are characterized by *constant* signs of the first and second derivatives. Clearly then, if we could automatically identify a sequence of segments S_{ij} over $[(t_i, x_i), (t_j, x_j)]$ with the first and second derivative signs constant, then the assignment of primitives would be straightforward. Relaxing the condition on the first derivative leads to an *unimodal* function. Figure 2 shows an example of data as a sequence of unimodal regions and the associated primitives.

Thus, our aim here is to extract out the *piecewise unimodals* parameterized as piecewise polynomi-

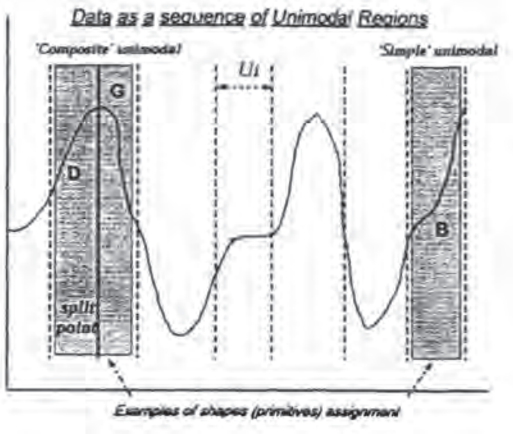

Fig. 2. Data as a sequence of *unimodals* regions

als. The problem of course lies in identifying these crucial segments ***automatically***. Polynomial-fit is employed to parameterize data and improvements are obtained by an interval reduction scheme rather than by increasing the polynomial-order. Preliminary work in this regard can be found in Paritosh (1999). The simplest and the most obvious choice for the approximating polynomial is the ***quadratic***, given the derivative conditions outlined above.

An effective trend identification procedure should be based on a simple yet powerful representation scheme. Before we present the main contributions of this work we briefly summarize the wavelet-based-denoising next. Wavelet analysis addresses many of the related issues in a favourable manner (Bakshi and Stephanopoulos, 1994) and presents a powerful denoising technique. To estimate the significance of the polynomial fit-error, the noise estimate from wavelet analysis is used.

3.1 *Wavelet Denoising*

The most important advantage of wavelet-based analysis is the advantage of minimal ***a priori*** information. Their excellent time-frequency localization allows good non-parametric statistical estimation of the ***true*** i.e., denoised data. Any process signal can be represented as

$$y(t) = f(t) + \epsilon(t) \tag{1}$$

where $y(t)$ = noisy data, $f(t)$ = unknown 'true' signal and $\epsilon(t)$ (noise) is usually assumed to be independent and identically distributed normal errors $\sim N(0, \sigma^2)$. The orthogonal wavelet series approximation to a continuous time signal $y(t)$ is given by

$$y(t) \approx \sum_k s_{J,k}\phi_{J,k}(t) + \sum_{j=1}^{j=J}\sum_k d_{j,k}\psi_{j,k}(t) \tag{2}$$

where J is the number of multiresolution components (or ***scales***), and k is the number of coefficients at the level. $s_{J,k}$ are called smooth coefficients and $d_{J,k}, \ldots, d_{1,k}$ are the detail coefficients. The functions $\phi_{j,k}(t)$ and $\psi_{j,k}(t)$ are approximating wavelet functions generated from ϕ and ψ by scaling (factor 2^j) and translation (parameter $2^j k$). When a discrete signal $y_1, y_2, \ldots, y_n$ (sampled data) is used, this approximation is referred to as the Discrete Wavelet Transform (DWT).

There exists an impressive theory for nonparametric regression and smoothing based on wavelet shrinkage (Donoho and Johnstone, 1994). The principle consists of (1) applying the DWT, (2) shrinking the coefficients $\to 0$ and (3) applying the inverse discrete wavelet transform (IDWT). The shrinking of wavelet coefficients usually is defined by the shrinkage function $\delta_c(x)$, (c = shrinkage threshold). The threshold is given usually as $c_j = \lambda_j\sigma_j$ where λ_j is the ***threshold rule***, σ_j is the ***noise scale***. Usually, λ is taken to be the universal threshold defined by $\sqrt{2\log N}$ (N = sample size) and results in high degree of smoothness. In cases of white noise, the finest scale detail coefficients $\mathbf{d_1}$ are used to estimate a single scale factor for all levels whereas for nonwhite noise, a level-dependent scale factor is estimated as $\sigma_j = \hat{\sigma}(\mathbf{d_j})$. The $\hat{\sigma}$ is the Median Absolute Deviation (MAD) function, a highly robust estimate of scale. The estimated sensor noise (σ_{noise}) and Signal/Noise ratio (S/N) can then be simply estimated as:

$$\sigma_{noise} = \sigma(y(t) - \hat{f}(t)) \tag{3}$$
$$S/N = \frac{\sigma_{\hat{f}}}{\sigma_{noise}}$$

where $\hat{f}(t)$ is the wavelet estimated denoised function of f. The next section discusses the proposed trend extraction technique.

4. INTERVAL-HALVING BASED TREND IDENTIFICATION

Any $y(t)$ can be represented arbitrarily closely using a sequence of piecewise polynomials $p_i(t)$ over the unimodal regions U_i each of order N_i i.e., $y(t) \approx \{p_1(t), p_2(t), \ldots, p_n(t)\}$. Thus the problem of identification of U_i is equivalently posed as the ***location*** and ***specification*** (coefficients) of these p_i. The algorithm consists of two parts: (1) determining the sequence of the p_i i.e.,U_i using the interval-halving procedure, and (2) assigning ***primitives*** to these identified U_i based on the signs of the derivatives of p_i. The standard least-squares result is used to fit a polynomial p_i with coefficients $\hat{\beta}_i$ to the data:

$$y = T\hat{\beta} \tag{4}$$

$$p_i = \sum_{n=0}^{n=2} \hat{\beta}_n t^n \text{ where } \hat{\beta} = (T^T T)^{-1} T^T y$$

where the i^{th} row of the T matrix $= [1 \cdots t^n]$, where n ($n < 2$) is the order of p_i. We now discuss the details of both the parts below:

4.1 *Polynomial Identification of Unimodals*

Let the data $y(t)$ be given as a sequence of sampled data points $y = (y_1, y_2, \ldots, y_N)$. Set start time $T_i = 1$ and end time $T_f = N$. Normalize the window of identification $W_{id} = [T_i\ T_f]$ to $[0\ 1]$. Set the initial window length $l = N$ and polynomial order $n = 0$.

(1) *Polynomial Fit:* Fit polynomial of order n, p_n to y in normalized $[T_i\ T_f]$ i.e., $[0\ 1]$. Calculate fit error ϵ^2_{fit} as

$$\epsilon^2_{fit} = \frac{1}{\nu_1} \sum_{i=1}^{i=l} (y_i - p_{n_i})^2 \qquad (5)$$

where ν_1 is the degrees of freedom $= l - (n+1)$. To determine the 'goodness of fit' i.e., significance of error we use the estimate of noise provided by the wavelet analysis in Section 3.1. The significance of ϵ^2_{fit} is tested against the wavelet noise estimate σ^2_{noise} using the following test:
F-test: Consider the random variable $F(\nu_1, \nu_2) = \frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2}$, where $s_1^2 = \epsilon^2_{fit}, s_2^2 = \sigma^2_{noise}$ and σ_1^2, σ_2^2 denote the population variances from which these sample variances are estimated and $\nu_2 = (N-1)$. We assume $\sigma_1^2 = \sigma_2^2$, thus $F(\nu_1, \nu_2) = \frac{s_1^2}{s_2^2}$. The statistical hypotheses:

$$\begin{aligned} H_0 &: \epsilon^2_{fit} = \sigma^2_{noise} \\ H_1 &: \epsilon^2_{fit} > \sigma^2_{noise} \end{aligned} \qquad (6)$$

are then evaluated using an one-sided *F-test* at a level of significance α i.e.,

$$\frac{\epsilon^2_{fit}}{\sigma^2_{noise}} < F_{1-\alpha}[\nu_1, \nu_2] \qquad (7)$$

If the inequality is satisfied, H_0 is accepted. If $T_f = N$ then the whole data record length N of y is covered, go to Step 3. Else the error is significant and move to Step 2.

(2) *Interval-Halving Step:* If the length l of W_{id} is below a threshold W_{thresh}, stop reducing the current segment and go to Step 1 with the left segment i.e., $[T_f\ T_N]$. The rationale behind W_{thresh} (say 10) is that if the window does fall to such a small number, we go ahead with the current polynomial fit and don't try to refine the interval any further.

If $n < 2$, increase $n = n + 1$ and repeat Step 1. If $n = 2$, *split* the current interval $W_{id} = [T_i\ T_f]$ at the *midpoint* (i.e., halve the interval) and assign $T_{half} = T_f/2$, $W_{id} = [T_i\ T_{half}]$. Go to Step 1 with the data segment in $[T_i\ T_{half}]$.

(3) The data y is thus been completely parameterized as a sequence of piecewise quadratics p_i i.e., $f = \bigcup_{i=1}^{M} p_i$ i.e., M unimodal regions U_i have been extracted. Quit.

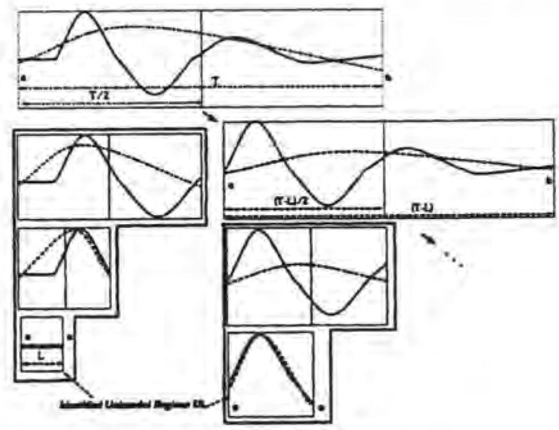

Fig. 3. Illustration of Interval-Halving

Figure 3 illustrates the idea: data length is recursively halved until an unimodal region (quadratic) with acceptable error is found. This process will terminate within a finite number of iterations K on an interval I. This is because there always exists an interval $[a\ b] \subset I$ which will be encountered wherein either (a) the *F-test* would succeed or (b) window length falls below W_{thresh} which would forcibly terminate the search.

4.2 *Assignment of Qualitative Shapes*

The trend language of *primitives* consisting of 7 letters is used here i.e., **A**(0,0), **B**(+,+), **C**(+,0), **D**(+,-), **E**(-,+), **F**(-,0), **G**(-,-) (Figure 1). The second part comprises of assigning primitives to the the piecewise polynomials and the labeling is based on the sign of the first ($d1_t$) and second ($d2_t$) derivatives. To examine the significance of the end-derivatives i.e., whether $d1_{t=0,1} \approx 0$ because of noise etc. a statistical test is carried out on the coefficients $\hat{\beta}$ with variance s_i^2.

(1) *Significance of derivatives:* Note that the second derivative $d^2p_i/dt^2 = 2\hat{\beta}_2$ is constant, while the first derivative $dp_i/dt = \hat{\beta}_1 + 2\hat{\beta}_2 t$ is a function of t. Calculate the first derivatives at the end-points $d1_0 \equiv dp_i/dt|_{t=0} = \hat{\beta}_1$ and $d1_1 \equiv dp_i/dt|_{t=1} = \hat{\beta}_1 + 2\hat{\beta}_2$. To examine the significance of the coefficients from the viewpoint of derivative calculation, since

derivatives would rarely be *identically zero* the following *T-test* is carried out.

T-test : The least squares estimate $\hat{\beta}$ is unbiased i.e., $E(\hat{\beta}) = \beta$ where $y = T\beta + e$. Also $covar(\hat{\beta}) = (T^T T)^{-1} ee^T$. We approximate ee^T with σ^2_{noise}. Next a two-sided *t-test* is done to bound β as follows:

$$\underbrace{\hat{\beta}_i - t_{1-\frac{\alpha}{2}} s_i}_{\hat{\beta}_{low}} < \beta_i < \underbrace{\hat{\beta}_i + t_{1-\frac{\alpha}{2}} s_i}_{\hat{\beta}_{high}} \quad (8)$$

where s_i is the standard deviation of $\hat{\beta}_i$ i.e., the i^{th} diagonal element of $\sqrt{(T'T)^{-1}\sigma^2_{noise}}$. Once the low and high limits are placed for each of the β, bounds are placed on the first derivatives at the ends (t = 0,1) as:

$$\underbrace{\hat{\beta}_{low_1} + 2\hat{\beta}_{low_2} t}_{deriv_{low}} < \frac{dp_i}{dt} < \underbrace{\hat{\beta}_{high_1} + 2\hat{\beta}_{high_2} t}_{deriv_{high}} (9)$$

At each end, if the minimum and maximum values of the derivatives i.e., $deriv_{low}$ and $deriv_{high}$ respectively are of ***opposite signs*** then the corresponding derivative is zeroed. If only one end derivative is zeroed, then the sign of the derivative at the other end (if not zero) is assigned to it. This procedure is carried out for all the M identified regions and thus new set of first derivatives $d1new_t$ are assigned to the end points ($t = 0, 1$).

(2) ***Assignment of Primitives:*** For $i = 1, \ldots, M$ check if the signs of the end derivatives in U_i match i.e., $sign(d1new_0) = sign(d1new_1)$. If yes, we have a 'simple' unimodal region for this section. If not, we have a composite region implying that the identified quadratic p_i in U_i is composed of 2 primitives (**EB** or **DG**). In that case find the zero-derivative point i.e., $\{t_{d1=0} : dp_i/dt|_{t=t_{d1=0}} = 0\}$ and split the composite U_i into 2 simple regions $\{[0\ t_{d1=0}],[t_{d1=0}\ 1]\}$. Thus we will have a sequence of simple unimodal regions and the primitive assignment is straightforward based on the derivative signs.

The next section describes the application of this technique to simulated and real data.

5. TREND EXTRACTION EXAMPLES

In this section we demonstrate the application of the above algorithm on a variety of simulated and real industrial data to evaluate its efficacy. To estimate the noise (Section 3.1) in all the signals the 'db3' wavelet (order 3 from the daubechies orthogonal wavelet family) is used. Also the significance level $\alpha = 0.05$ for all the statistical tests.

5.1 *Performance Metrics*

To evaluate the effectiveness of the algorithm we use the following measures as performance metrics:

(1) Scaled Average Global Error (SAGE): This scaled error is calculated between between the wavelet estimated *denoised* data $\hat{f}$ and the piecewise quadratic approximation to the data $f_{approx} = \bigcup_{i=1}^{n} p_i$.

$$\text{SAGE} = \frac{1}{y_{range}} \sqrt{\frac{\sum (f_{approx} - \hat{f})^2}{N}}$$

(2) Scaled L_∞ error (SLE): Defined as the maximum local absolute error between the $\hat{f}$ and f_{approx}:

$$\text{SLE} = \frac{1}{y_{range}} \max |f_{approx} - \hat{f}|$$

(3) Compression Ratio (ρ): As a bonus of the piecewise quadratic representation, we also achieve data compression since the data is now parameterized in terms of the coefficients of the polynomials. For a data record of length N, if the approximation contains M piecewise segments and the number of nonzero coefficients in each is n_i then this ratio is defined as:

$$\rho = \frac{N}{\sum_{i=1}^{M} n_i + M + 1}$$

where the $M + 1$ in the denominator is for the number of end time points information required.

Though these metrics give an idea of the 'fit-performance' they aren't entirely suited for representing 'temporal performance', since a single number can't condense all information. For this reason we tested the method on a wide range of signals to test it exhaustively.

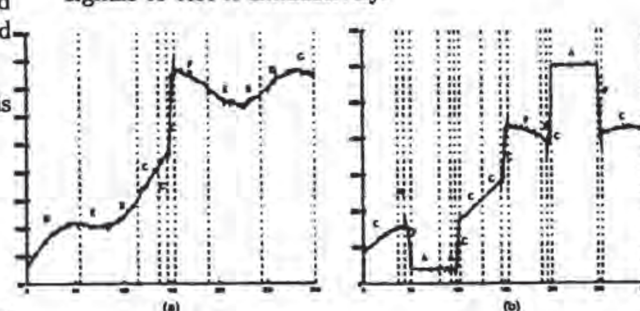

Fig. 4. Trends Extracted from Simulated (a)Case 2 (b)Case 3

Synthetic Data: The wavelet estimated noise is seen to be close to the simulated noise. Figure 4 shows the extracted trends for Cases 2 and 3. It is

Table 1. Signals and Performance

Case	Functional Form	Data Length	Simul. Noise	Est σ_{noise}	y_{range}	SAGE	SLE	ρ
		Simulated Signals						
1	$y = \begin{cases} 80 + 1.2t & \text{if } t \leq 150 \\ 400 & \text{if } 150 < t \leq 300 \end{cases}$	300	4	4.21	340.5	0.023	0.178	10.4
2	$y = \text{Case4} \; \& 30 sin(\frac{5}{100}t)$	300	4	4.30	355.7	0.019	0.1704	10.4
3	$y = \text{Case5} \; \& \begin{cases} 40 & \text{if } 50 \leq t \leq 100 \\ 600 & \text{if } 200 \leq t \leq 250 \end{cases}$	300	4	5.89	581.2	0.029	0.2012	6.5
		Industrial Signals						
1	Real	1441	–	2.1379	55.45	0.0406	0.1669	15.84
2	Real	1441	–	0.3463	16.97	0.02	0.0717	27.2
3	Real	1441	–	1.1011	29.18	0.05	0.2216	7.71

apparent from all these cases that the method performs very well in extracting all crucial features and matches 'desired' trends closely. Compression ratios are seen to vary between 6-17.

Industrial Data: In general these signals are more 'rough' compared to the smooth simulated signals above. Since the noise is not known *a priori*, level-dependent estimation of noise scale (Section 3.1) is used. Figure 5 shows trend extraction from two industrial cases. Compression ratios are seen to be as high as 14-21.

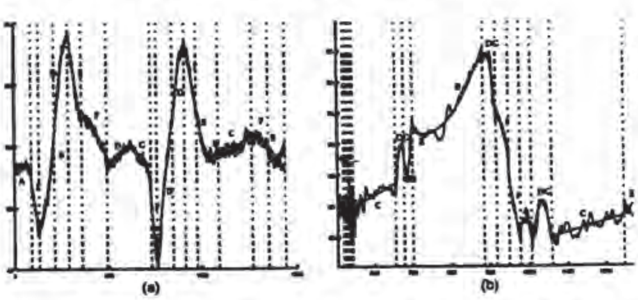

Fig. 5. Trends Extracted from Industrial (a) Case 2 (b) Case 3

The performance measures for all these signals are reported in Table 1.

6. CONCLUSIONS

The main issues in process trends analysis were discussed. A novel technique interval-halving technique was proposed which is both transparent and intuitive. The method generates a sequence of shapes (A-G) describing the data *qualitatively* and piecewise quadratics q_i explaining it *quantitatively* in a succinct manner. Data compression and denoising were added benefits as a result of polynomial smoothing and statistical testing. It is also *adaptive* to the scales of trend evolution and *robust* to process noise. We demonstrate the effectiveness of this technique on simulated and real plant data.

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